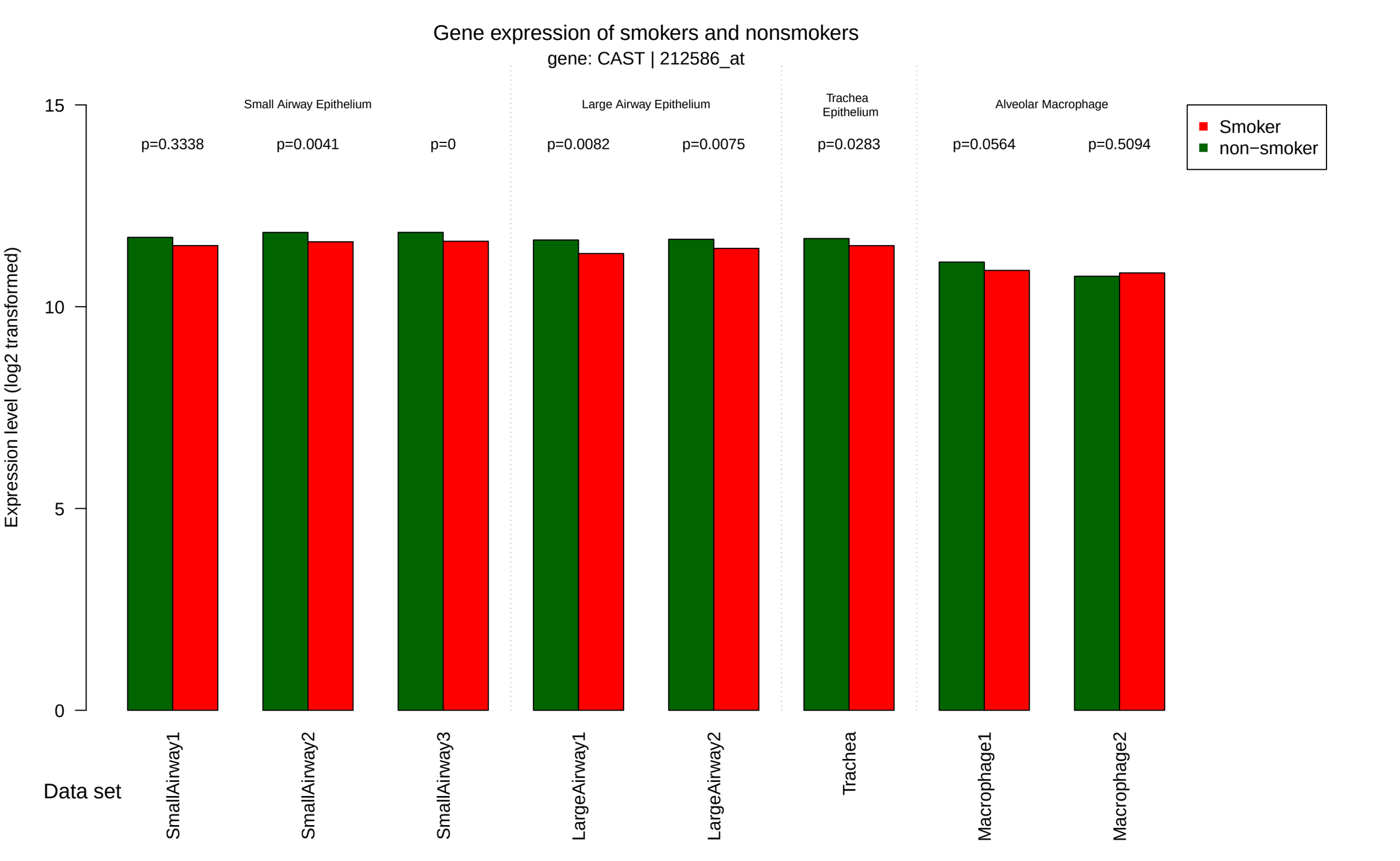
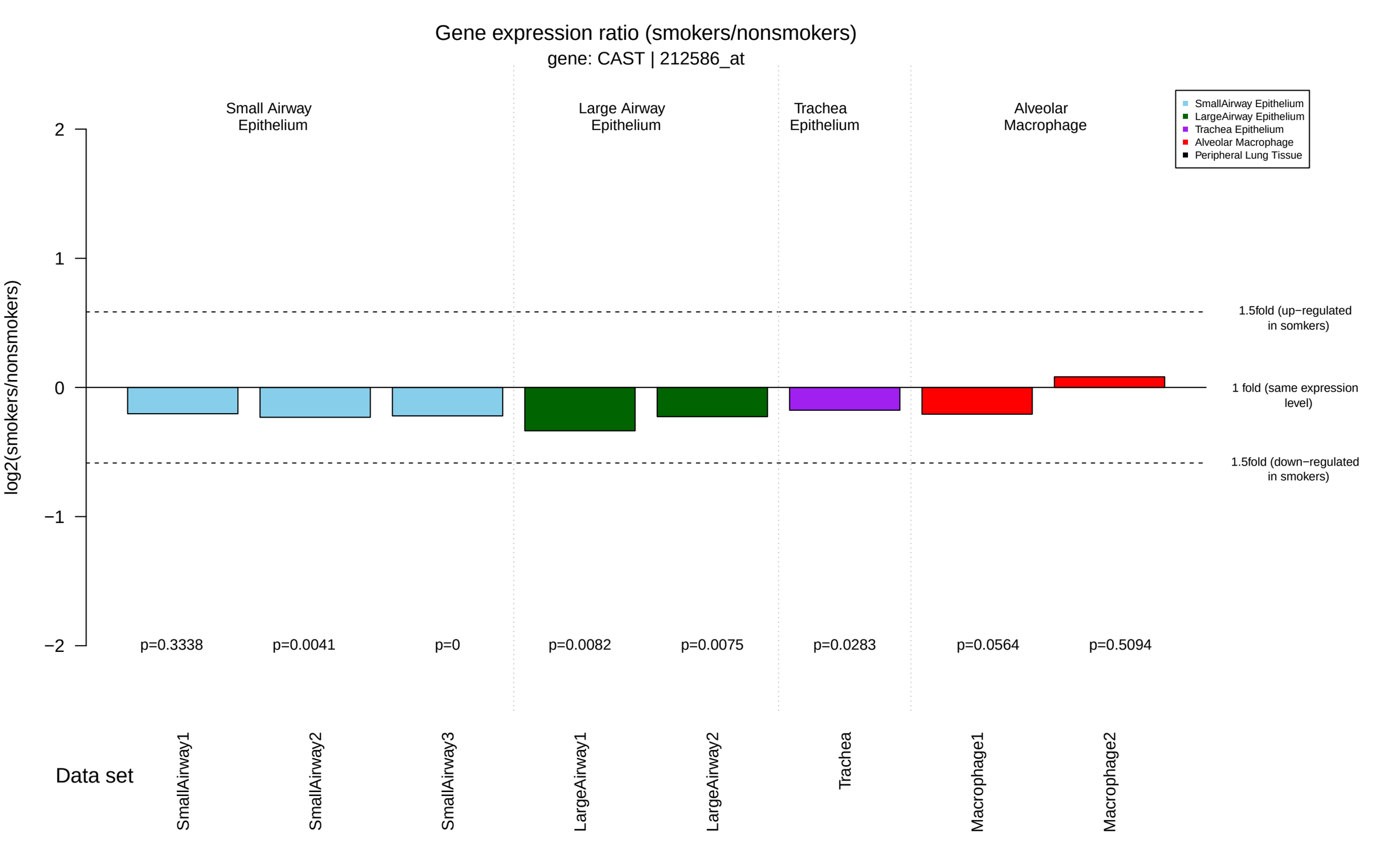
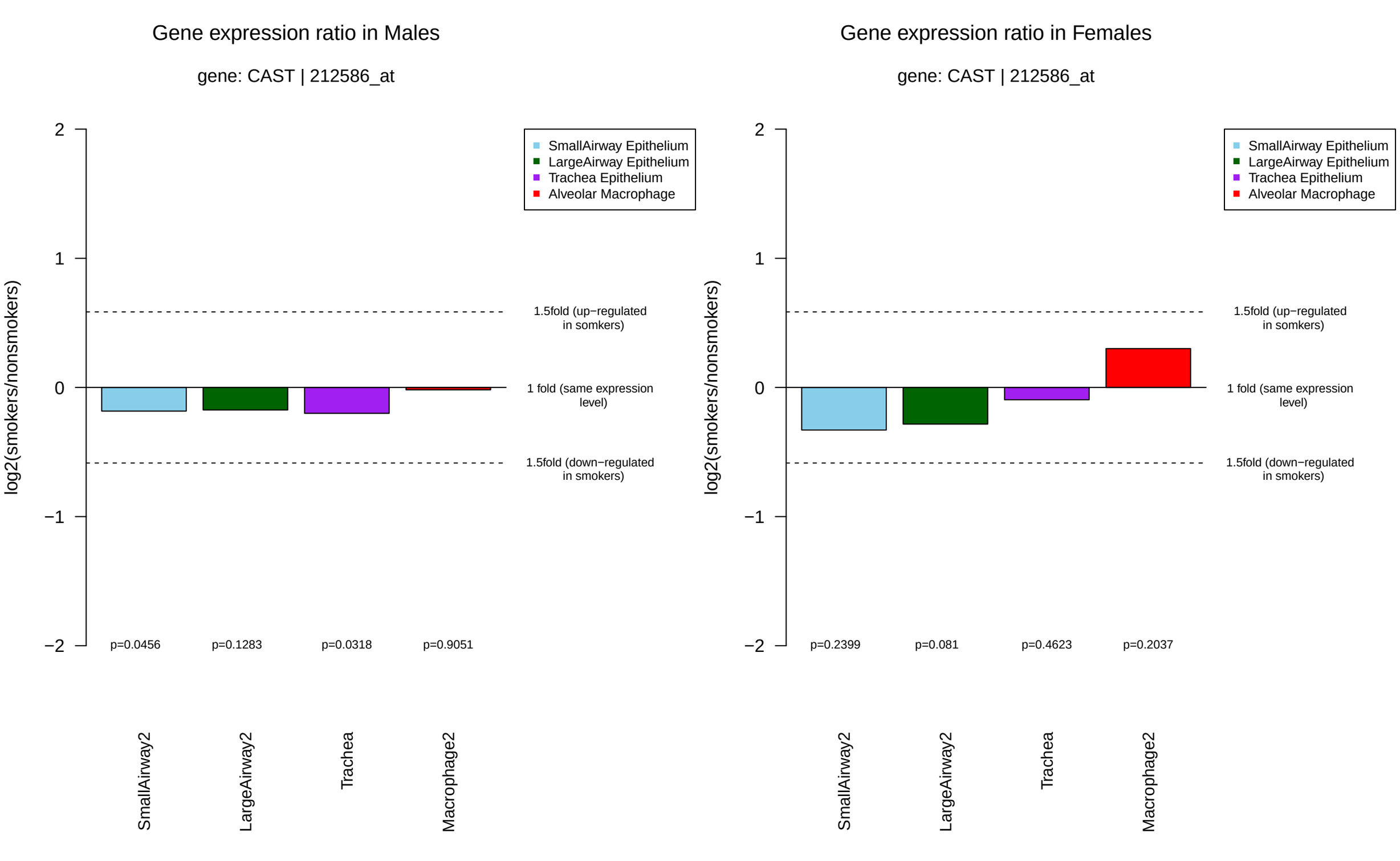
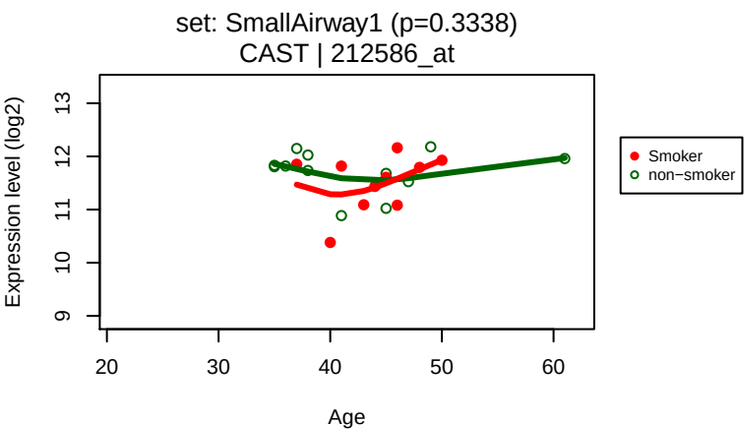
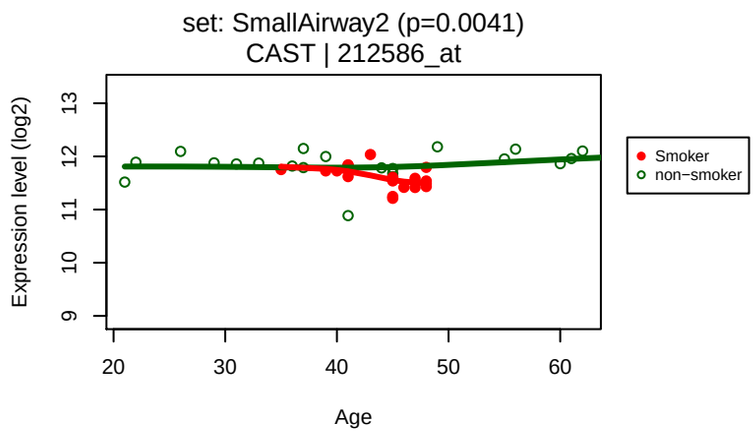
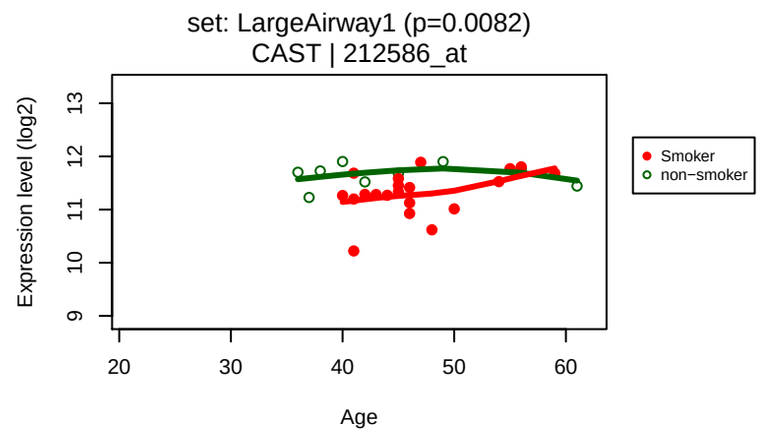

Gene expression of smokers and nonsmokers
gene: CAST | 212586_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3338
p=0.0041
p=0
p=0.0082
p=0.0075
p=0.0283
p=0.0564
p=0.5094
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

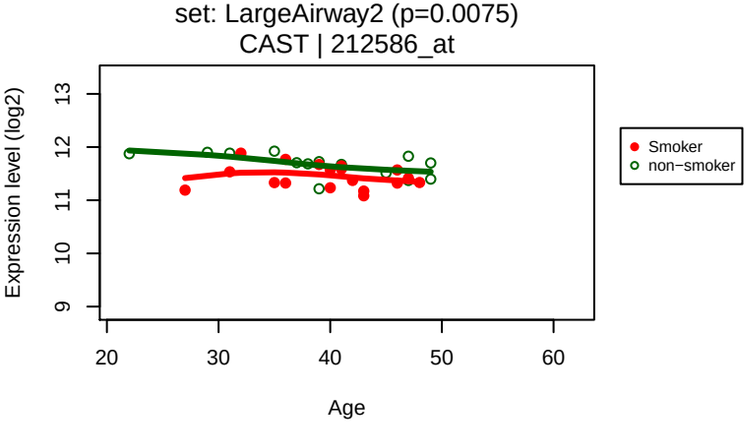

Gene expression ratio (smokers/nonsmokers)
gene: CAST | 212586_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3338
p=0.0041
p=0
p=0.0082
p=0.0075
p=0.0283
p=0.0564
p=0.5094
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

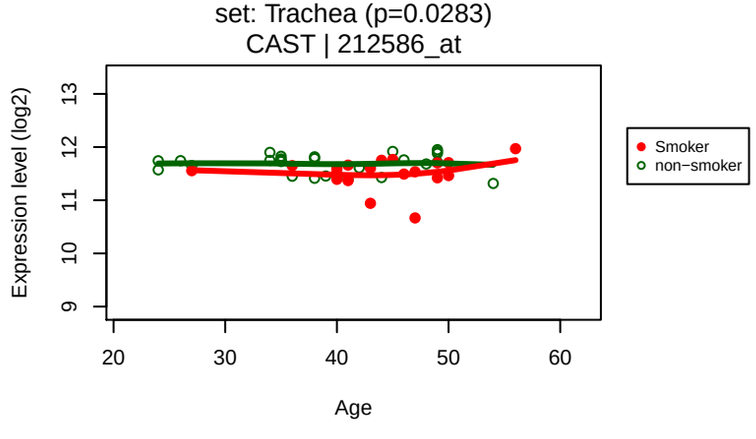

Gene expression ratio in Males
gene: CAST | 212586_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0456
p=0.1283
p=0.0318
p=0.9051
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CAST | 212586_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2399
p=0.081
p=0.4623
p=0.2037
SmallAirway2
LargeAirway2
Trachea
Macrophage2

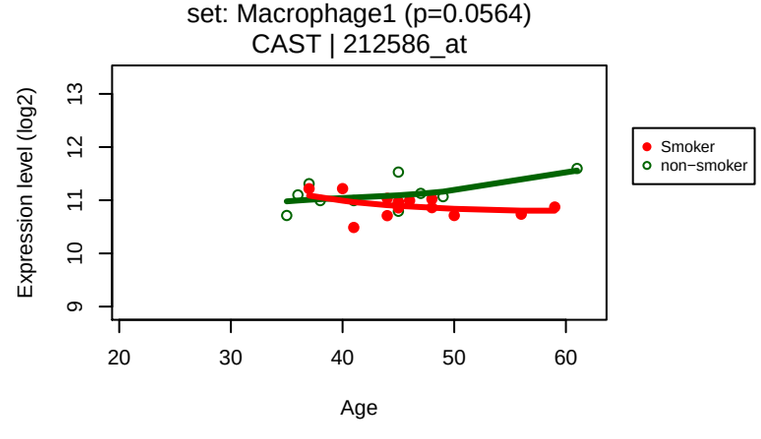

set: SmallAirway1 (p=0.3338)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker

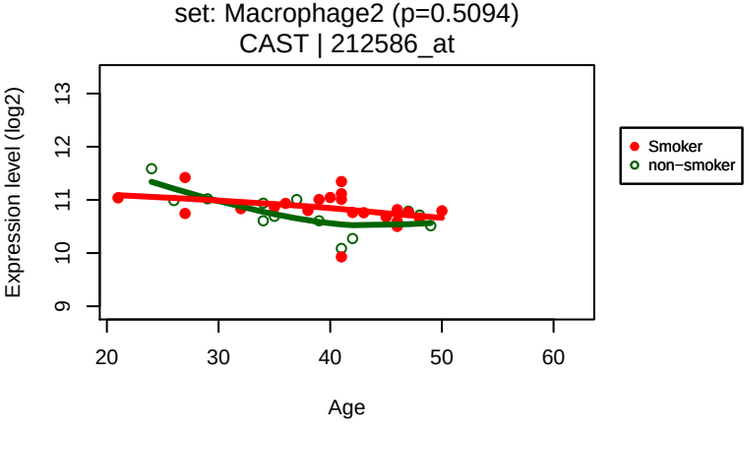

set: SmallAirway2 (p=0.0041)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker

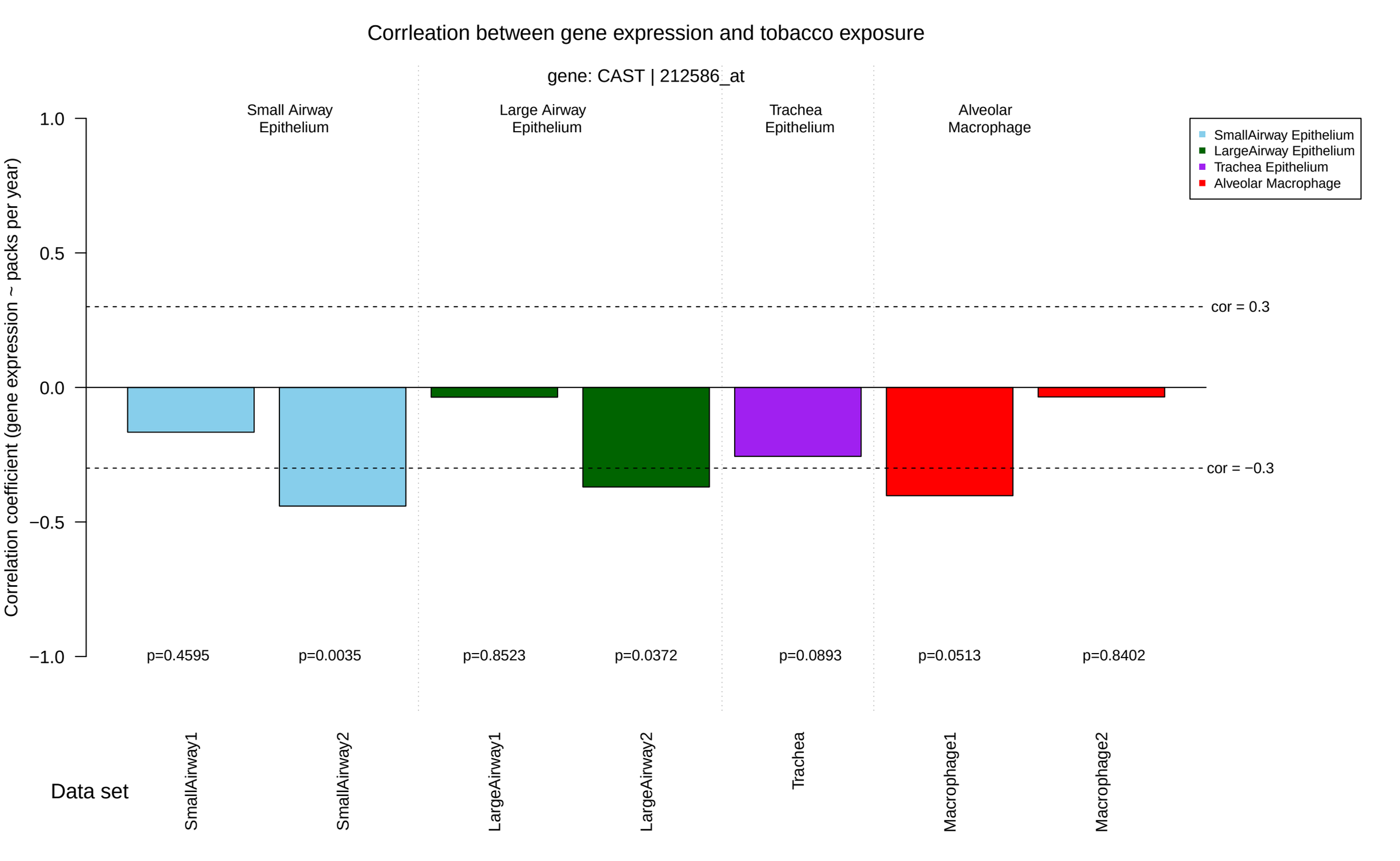

set: LargeAirway1 (p=0.0082)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0075)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0283)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0564)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5094)
CAST | 212586_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CAST | 212586_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4595
p=0.0035
p=0.8523
p=0.0372
p=0.0893
p=0.0513
p=0.8402
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2